

A Genus-Four Second Moment for the Fourier–Cubic Hénon Kernel

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Abstract

We identify the Galois-normalized second chronological moment of a Fourier–cubic Hénon kernel with the Frobenius trace of an explicit smooth genus-four curve. Projective direction counting reduces the four-variable zero fibre to a $(2, 3)$ complete intersection, which becomes a smooth $(3, 3)$ divisor on a split quadric. If a_p is its Frobenius trace, the descended moment is exactly $C_{p,2} = -14 - 2a_p$. The Weil bound supplies a square-root gain after field-degree normalization. It extends the associated Euler germ and its regularized graded operator determinant from $\operatorname{Re} s > 1/2$ to $\operatorname{Re} s > 1/3$; the operator formula now uses a sixth-order determinant relative to the normalized semifinite trace and five explicit chronological counterterms. This is not a classical Fredholm determinant on that half-plane: unregularized semifinite trace class begins at $\operatorname{Re} s > 2$, and classical Hilbert trace class begins at $\operatorname{Re} s > 3$ and encodes the ordinary Galois norm. No continuation, functional equation, or Riemann-divisor identification is claimed.

1 Introduction

The full Fourier–cubic quantization of the homogeneous area-preserving Hénon map produces exact finite-field trace moments. Pairing the additive character with its inverse and averaging over the maximal real cyclotomic field yields rational logarithmic moments $c_{p,n}$. Earlier work gives a normally convergent Euler germ on $\operatorname{Re} s > 1/2$ and realizes it as a fourth-order regularized graded determinant relative to a normalized semifinite trace. That threshold comes from using a uniform bound for every $n \geq 2$.

Here the second moment is evaluated geometrically. The main result is the following.

Theorem 1.1 (Genus-four second moment). *For every prime $p > 3$, $p \equiv 1 \pmod{3}$, the second descended Hénon moment has the form*

$$C_{p,2} = -14 - 2a_p, \quad |a_p| \leq 8\sqrt{p},$$

where $a_p = p + 1 - \#X_\rho(\mathbf{F}_p)$ is the Frobenius trace of an explicit smooth genus-four curve. Consequently the normalized Euler germ is holomorphic and nonzero on $\operatorname{Re} s > 1/3$, and on that half-plane

$$\mathcal{G}(s) = \exp\left(-\sum_{n=1}^5 \frac{\ell_n(s)}{n}\right) \operatorname{Det}_{6,\tau,\operatorname{gr}}(I - X_s).$$

Order six is the least fixed integer order in $L^q(\mathcal{M}, \tau)$ valid on the whole half-plane. More precisely,

$$X_s \in L^q(\mathcal{M}, \tau) \iff q \operatorname{Re} s > 2, \quad X_s \in S^q(\mathcal{H}) \iff q \operatorname{Re} s > 3.$$

Thus the unregularized τ -associated determinant begins at $\operatorname{Re} s > 2$, whereas the classical Fredholm domain begins at $\operatorname{Re} s > 3$. The latter realizes the ordinary Galois norm, not the field-degree-normalized root $\mathcal{G}$.

The geometric facts used below are classical: the split Fermat-cubic count [2, Cor. 8.3], split quadrics, adjunction on $\mathbf{P}^1 \times \mathbf{P}^1$ [1, Ch. V, Prop. 1.5], and the Weil bound for curves [4, Thm. 5.2.3]. The new content is the exact bridge from the frozen chronological Hénon phase to this curve and the resulting analytic abscissa. Regularized determinant background is recorded in [3, Chapter 9].

2 The chronological zero fibre

Fix an element $\rho \in \mathbf{F}_p^\times$ of order three. The four-step phase is

$$\Phi(x) = 2 \sum_{i=0}^3 x_i^3 + x_0x_1 + x_1x_2 + x_2x_3 + \rho x_3x_0. \quad (1)$$

Let $Z_p = \#\Phi^{-1}(0)$. The exact Galois trace normalization is

$$C_{p,2} = \frac{2Z_p}{p} - 2p^2, \quad c_{p,2} = \frac{2C_{p,2}}{p-1}. \quad (2)$$

Write $\mathcal{C} = \sum x_i^3$ and $\mathcal{Q} = x_0x_1 + x_1x_2 + x_2x_3 + \rho x_3x_0$, and denote by $S, R \subset \mathbf{P}^3$ their zero loci and by $X = S \cap R$.

Proposition 2.1 (Direction count). *One has*

$$Z_p = 1 + \#\mathbf{P}^3(\mathbf{F}_p) - \#S(\mathbf{F}_p) - \#R(\mathbf{F}_p) + p\#X(\mathbf{F}_p).$$

Proof. For a projective direction $([v])$,

$$\Phi(\lambda v) = \lambda^2(2\lambda\mathcal{C}(v) + \mathcal{Q}(v)).$$

If neither homogeneous value vanishes there is one nonzero root. If exactly one vanishes there is none. If both vanish all $p-1$ nonzero scalars work. Adding the origin proves the formula. $\square$

Because $\mu_3 \subset \mathbf{F}_p$, the Fermat cubic surface is split; its blow-up description gives

$$\#S(\mathbf{F}_p) = \#\mathbf{P}^2(\mathbf{F}_p) + 6p = p^2 + 7p + 1.$$

See also [2, Corollary 8.3]; since an odd prime congruent to one modulo three is congruent to one modulo six, its split case applies. The invertible change $y = x_1 + \rho x_3$, $w = x_1 + x_3$ turns $\mathcal{Q}$ into $x_0y + x_2w$. Hence R is a split quadric, $R \simeq \mathbf{P}^1 \times \mathbf{P}^1$, and $\#R = (p+1)^2$. Therefore

$$Z_p = p^3 - p^2 - 8p + p\#X(\mathbf{F}_p). \quad (3)$$

3 The genus-four intersection

The construction is uniform over $\mathcal{R} = \mathbf{Z}[\rho, 1/6]/(\rho^2 + \rho + 1)$. Since $N(\rho - 1) = 3$, the element $\rho - 1$ is a unit in $\mathcal{R}$. With $y = x_1 + \rho x_3$ and $w = x_1 + x_3$, the standard Segre parametrization of $x_0y + x_2w = 0$ is

$$x_0 = rt, \quad x_2 = ru, \quad y = -su, \quad w = st.$$

Equivalently,

$$x_0 = rt, \quad x_2 = ru, \quad x_1 = \frac{s(\rho t + u)}{\rho - 1}, \quad x_3 = -\frac{s(t + u)}{\rho - 1}.$$

Using $(\rho - 1)^2 = -3\rho$, direct substitution gives the exact identity $\rho\mathcal{C}|_R = F$, where

$$X_\rho : \quad \rho r^3(t^3 + u^3) + \rho^2 s^3 t^2 u - s^3 t u^2 = 0 \quad \subset \mathbf{P}^1 \times \mathbf{P}^1. \quad (4)$$

Proposition 3.1. *The curve X_ρ is smooth and geometrically irreducible for every allowed p , and it has genus four.*

Proof. Smoothness is proved uniformly in Appendix A. Equation (4) has bidegree $(3, 3)$. Over an algebraic closure, suppose its components could be partitioned into nonzero effective divisors of bidegrees (a, b) and $(3 - a, 3 - b)$. Components of a smooth curve are disjoint, but their intersection number is

$$a(3 - b) + b(3 - a) > 0$$

for every nontrivial partition. This contradiction proves geometric connectedness and hence geometric irreducibility. Adjunction [1, Chapter V, Section 1, Proposition 1.5 and Example 1.5.2] gives $g = (3 - 1)(3 - 1) = 4$. $\square$

Put $a_p = p + 1 - \#X_\rho(\mathbf{F}_p)$. Substitution of (3) into (2) proves

$$\boxed{C_{p,2} = -14 - 2a_p}, \quad \boxed{c_{p,2} = -\frac{28 + 4a_p}{p - 1}}. \quad (5)$$

The Weil bound now gives

$$|a_p| \leq 8\sqrt{p}, \quad |c_{p,2}| \leq \frac{28 + 32\sqrt{p}}{p - 1}. \quad (6)$$

The point count is independent of the choice between ρ and $\rho^{-1} = \rho^2$. Indeed, for

$$T([r : s], [t : u]) = ([r : -s], [u : t])$$

one checks directly that $F_{\rho^{-1}} = \rho F_\rho \circ T$. Thus T is an $\mathbf{F}_p$ -isomorphism between the two presentations.

4 The third abscissa and two trace categories

The logarithm of the normalized Euler product is

$$- \sum_{p \equiv 1(3)} \sum_{n \geq 1} \frac{c_{p,n}}{n} p^{-ns}.$$

The exact first moment $c_{p,1} = -12/(p - 1)$ is summable for $\operatorname{Re} s > 0$. By (6), the $n = 2$ contribution is summable for $\operatorname{Re} s > 1/4$. For $n \geq 3$, the inherited estimate $|c_{p,n}| \leq 4 \cdot 4^n$ gives the large-prime majorant, but a finite-prime step is needed to sum over n . Fix a compact set with $\operatorname{Re} s \geq \sigma_0 > 1/3$, and choose P_0 so that $4p^{-\sigma_0} \leq 1/2$ for $p > P_0$. Then

$$\sum_{p > P_0} \sum_{n \geq 3} \frac{|c_{p,n}|}{n} p^{-n\sigma_0} \ll \sum_{p > P_0} p^{-3\sigma_0} < \infty.$$

For the finitely many $p \leq P_0$, the unitary-block estimate $|c_{p,n}| \leq \tau_p(I) = (8p + 4)/3$ makes the logarithmic series converge geometrically because $p^{-\sigma_0} < 1$. This proves local normal convergence and the Euler assertion in Theorem 1.1.

The operator interpretation is equally exact, but the trace category must be specified. Let $(\mathcal{M}, \tau)$ be the inherited product algebra with its normalized faithful semifinite trace, and let X_s be the Galois-sector block. Since each local block is unitary,

$$\tau(|X_s|^q) = \sum_{p \equiv 1(3)} \frac{8p + 4}{3} p^{-q \operatorname{Re} s}, \quad X_s \in L^q(\mathcal{M}, \tau) \iff q \operatorname{Re} s > 2.$$

In particular, X_s is τ -trace class exactly for $\operatorname{Re} s > 2$; only there does the unregularized τ -associated analytic graded determinant exist. On the larger half-plane $\operatorname{Re} s > 1/3$, $L^6(\mathcal{M}, \tau)$ is the first fixed integer ideal covering every point. With

$$\ell_n(s) = \sum_p c_{p,n} p^{-ns}, \quad 1 \leq n \leq 5,$$

the $n = 1$ and $n = 2$ estimates above, together with the uniform bound for $3 \leq n \leq 5$, show that every counterterm converges there. Removing the first five logarithmic terms produces the trace-associated graded regularized determinant

$$\operatorname{Det}_{6,\tau,\operatorname{gr}}(I - X_s) = \exp \left(- \sum_{n \geq 6} \frac{\operatorname{str}(X_s^n)}{n} \right),$$

equivalently the quotient of the τ -associated sixth-order regularized determinants in the positive and negative grades [3, Chapter 9]. This gives the factorization in Theorem 1.1. Order six is the least fixed integer order forced by positive $L^q(\mathcal{M}, \tau)$ -membership on the whole half-plane; the claim does not classify unrelated regularization schemes.

The ordinary Hilbert trace gives a different threshold and a different local object. Let $\mathcal{H} = \bigoplus_p \mathcal{H}_p$ be the underlying Hilbert direct sum and $d_p = (p - 1)/2$. Since

$$\dim \mathcal{H}_p = d_p \tau_p(I) = \frac{(p - 1)(4p + 2)}{3},$$

one has

$$\operatorname{Tr}_{\mathcal{H}}(|X_s|^q) = \sum_{p \equiv 1(3)} \frac{(p - 1)(4p + 2)}{3} p^{-q \operatorname{Re} s}, \quad X_s \in S^q(\mathcal{H}) \iff q \operatorname{Re} s > 3.$$

At equality the series is a positive constant multiple of the divergent split-prime harmonic series; the two strict inequalities follow by the same prime-series comparison. Thus classical Hilbert trace class begins only at $\operatorname{Re} s > 3$. Furthermore,

$$\operatorname{Str}_{\mathcal{H}_p}(W_p^n) = d_p c_{p,n} = C_{p,n},$$

Writing G_p for the local normalized root and $N_p = G_p^{d_p}$ for its ordinary Galois norm, the canonical local logarithm satisfies

$$\exp \operatorname{Str}_{\mathcal{H}_p} \operatorname{Log}_0(I - zW_p) = G_p(z)^{d_p} = N_p(z).$$

The classical Fredholm determinant therefore encodes the ordinary Galois norm N_p , rather than the normalized root G_p used in $\mathcal{G}$. The larger regularization order in the normalized category is the transparent price of reaching a larger half-plane; it is not a loss of Hénon chronology.

5 Route-A audit and the next wall

The formal evaluation is

$$\begin{aligned} & \text{A1_WEAK,} \quad \text{A2_ANALYTIC_DETERMINANT,} \\ & \text{A3_PARTIAL_ANALYTIC_STRUCTURE,} \quad \text{A4_NATURAL_QUANTIZATION.} \end{aligned}$$

The advance is exact: a chronological Hénon moment has become a Frobenius trace, the Euler germ reaches $\operatorname{Re} s > 1/3$, and a canonical regularized operator determinant relative to the normalized semifinite trace realizes it. It is neither an unregularized τ -determinant there—that

domain starts at $\operatorname{Re} s > 2$ —nor a classical Fredholm determinant, whose trace-class domain starts at $\operatorname{Re} s > 3$ and whose local trace gives the ordinary norm. A1 remains weak because primes label finite arithmetic fibres rather than primitive cycles of a single real map. A3 remains partial because there is no continuation, Gamma factor, functional equation, zero-counting law, or Riemann-divisor match. A4 is natural at each finite place, but no global self-adjoint spectral generator has been built.

The first unresolved logarithmic term is now $n = 3$. Its direction stratification involves a cubic fourfold, a split quadric fourfold, and their $(2, 3)$ threefold intersection. Isolating the non-Tate middle cohomology and proving a second square-root gain is the next decisive gate; absent that gain, $\operatorname{Re} s = 1/3$ remains the certified wall.

A Uniform smoothness

We give a literal four-chart Jacobian certificate. Work over an algebraic closure of a field of characteristic different from two and three. On the charts $U_{rt}, U_{ru}, U_{st}, U_{su}$, use affine coordinates $(\xi, \eta) = (s, u), (s, t), (r, u), (r, t)$, respectively. Equation (4) becomes

$$\begin{aligned} f_{rt} &= \rho(1 + \eta^3) + \xi^3\eta(\rho^2 - \eta), \\ f_{ru} &= \rho(\eta^3 + 1) + \xi^3\eta(\rho^2\eta - 1), \\ f_{st} &= \rho\xi^3(1 + \eta^3) + \eta(\rho^2 - \eta), \\ f_{su} &= \rho\xi^3(\eta^3 + 1) + \eta(\rho^2\eta - 1). \end{aligned}$$

On U_{rt} , the equation $\partial_\xi f_{rt} = 3\xi^2\eta(\rho^2 - \eta) = 0$ leaves $\xi = 0$, $\eta = 0$, or $\eta = \rho^2$. If $\xi = 0$, then $\partial_\eta f_{rt} = 3\rho\eta^2 = 0$, hence $\eta = 0$, but $f_{rt} = \rho$. The other two cases give $f_{rt} = \rho$ and $f_{rt} = 2\rho$, respectively. Thus this chart has no singular point.

On U_{ru} , the equation $\partial_\xi f_{ru} = 3\xi^2\eta(\rho^2\eta - 1) = 0$ leaves $\xi = 0$, $\eta = 0$, or $\eta = \rho$. The first case, together with $\partial_\eta f_{ru} = 0$, reduces to $\eta = 0$ and $f_{ru} = \rho$; the remaining cases give $f_{ru} = \rho$ and $f_{ru} = 2\rho$. Again there is no singular point.

On U_{st} , the equation $\partial_\xi f_{st} = 3\rho\xi^2(1 + \eta^3) = 0$ leaves $\xi = 0$ or $\eta^3 = -1$. In the first case, $f_{st} = 0$ forces $\eta = 0$ or $\eta = \rho^2$, whereas $\partial_\eta f_{st}$ is respectively ρ^2 or $-\rho^2$. In the second case, $f_{st} = 0$ forces $\eta = \rho^2$, whose cube is one, contradicting characteristic different from two.

Finally, on U_{su} , the equation $\partial_\xi f_{su} = 3\rho\xi^2(1 + \eta^3) = 0$ leaves $\xi = 0$ or $\eta^3 = -1$. If $\xi = 0$, then $f_{su} = 0$ forces $\eta = 0$ or $\eta = \rho$, where $\partial_\eta f_{su}$ equals -1 or 1 . If $\eta^3 = -1$, then $f_{su} = 0$ forces $\eta = \rho$, whose cube is one, another contradiction.

The four charts cover $\mathbf{P}^1 \times \mathbf{P}^1$, so the curve is geometrically smooth in every characteristic other than two and three. Together with $N(\rho - 1) = 3$, this is an all-prime certificate whose only excluded denominators are supported on two and three; no numerical Jacobian test is used.

References

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